EXPERIMENT NO: 17

Experiment to understand Linear Regression for a given data set.

PROGRAM:

```
import numpy as np
import pandas as pd
df=pd.read_csv('Salary_data.csv')
print(df.info())
df.dropna(inplace=True)
print(df.info())
print(df.describe())
features=df.iloc[:,[0]].values
label=df.iloc[:,[1]].values
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(features, label, test_size=0.2,random_state=42)
from sklearn.linear_model import LinearRegression
model=LinearRegression()
model.fit(x_train,y_train)
print(model.score(x_train,y_train))
print(model.score(x_test,y_test))
print(model.coef_)
print(model.intercept_)
import pickle
pickle.dump(model,open('SalaryPred.model','wb'))
model=pickle.load(open('SalaryPred.model','rb'))
yr_of_exp=float(input("Enter Years of Experience: "))
yr_of_exp_NP=np.array([[yr_of_exp]])
Salary=model.predict(yr_of_exp_NP)
print("Estimated Salary for {} years of experience is {}: ".format(yr_of_exp,Salary))
```

OUTPUT:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 2 columns):
 #   Column           Non-Null Count  Dtype  
---  ------           --------------  -----  
 0   YearsExperience  30 non-null     float64
 1   Salary           30 non-null     int64  
dtypes: float64(1), int64(1)
memory usage: 612.0 bytes
```

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```

	YearsExperience	Salary
count	30.000000	30.000000
mean	5.313333	76003.000000
std	2.837888	27414.429785
min	1.100000	37731.000000
25%	3.200000	56720.750000
50%	4.700000	65237.000000
75%	7.700000	100544.750000
max	10.500000	122391.000000

LinearRegression

LinearRegression()

0.9024461774180497 array([[9423.81532303]]) array([25321.58301178])

Estimated Salary for 44.0 years of experience is [[439969.45722514]]:

Enter Years of Experience: 44